

A Mathematical Backbone for MMALS:

From Mycelial Route Intuition to Distributional Memory, Information Geometry, and
Context-Aware Continual Learning

MMALS Project Research Note¹

¹Internal arXiv-style draft for hypothesis preservation and research planning

May 2026

Abstract

Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS) were initially motivated by a biological analogy: multiple specialist modules, interpreted as host organisms, exchange transformed latent information through a fungal medium that routes, charges, reinforces, prunes, and preserves useful pathways. Early prototypes established a critical diagnostic distinction: preserving the selected route is not equivalent to preserving the computation performed through that route. This note formalizes the “mathematical backbone” behind that distinction. It argues that MMALS should evolve from route-vector memory toward distributional functional memory, geometric routing, and uncertainty-aware context inference. The first pillar is Fréchet’s 1957 analysis of distances between probability laws, which motivates comparing memory distributions rather than isolated route vectors. The second pillar is optimal transport, interpreted as the cost of transforming one latent memory law into another. The third pillar is information geometry, especially Fisher-type metrics and Fréchet–Darmois/Cramér–Rao bounds, which provide a principled geometry for routing on probability simplexes and a limit theory for inferred contexts. The fourth pillar is a future quantum-inspired extension in which functional memory is represented as a positive semidefinite state over route-function compositions. The paper keeps the 12-year-old intuition—forest, tunnels, jars of marbles, sand piles, and probability playgrounds—but translates it into a PhD-level research program with definitions, propositions, falsification criteria, and an experimental roadmap.

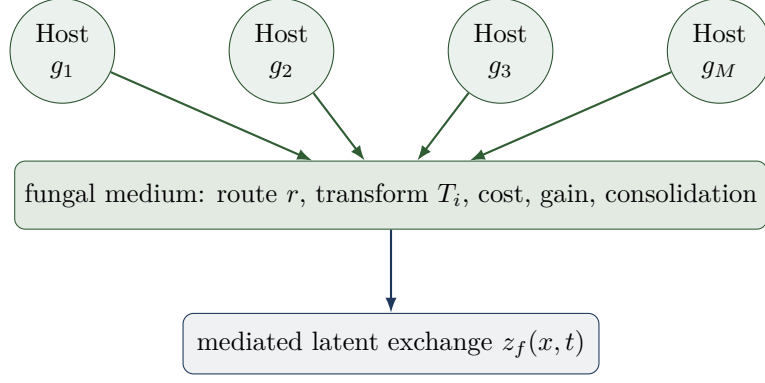
Keywords: continual learning; catastrophic forgetting; mycelium-inspired learning; functional memory; Fréchet distance; optimal transport; Fisher geometry; natural gradient; information geometry; quantum-inspired machine learning.

Status. This document is an internal research note. It is not a publication claim, not a proof of MMALS superiority, and not a claim of physical quantum advantage. It is a formalization of the conceptual and mathematical discussion surrounding the MMALS roadmap.

1 Motivation: from a forest metaphor to a research object

A child-friendly explanation of MMALS starts with a forest. Each tree is a specialist learner. The underground mycelium is not a passive cable: it decides which tree should share information, transforms what is exchanged, and pays a “carbon” cost when communication is too diffuse. A standard neural network resembles a single student asked to remember everything; MMALS resembles a forest school in which multiple specialists collaborate through an adaptive communication substrate.

The metaphor is useful only if it generates falsifiable mathematical objects. The MMALS v0.x program progressively converted the metaphor into a sequence of experimental claims: structural routing, route sparsity, metabolic cost, mutualistic gain, soft consolidation, output distillation, latent distillation, and finally route-function disentanglement. The v0.9 diagnostic



The mathematical question is not only which tunnel is used, but whether the meaning transported through the tunnel is preserved.

Figure 1: Forest-level intuition and formal MMALS objects. Hosts emit representations, the fungal medium routes and transforms them, and the mediated latent exchange becomes the memory-relevant computation.

notebook framed the central reviewer-level question: is memory in MMALS the route itself, or the function computed through the route? The project answer was clear: route topology alone is not memory; the relevant object is the mediated latent function computed through the route [6].

The purpose of this paper is to give that claim a mathematical backbone. A backbone is not a decoration. It must answer four questions:

1. What is the memory object?
2. What metric tells us that memory has drifted?
3. What geometry should constrain route updates?
4. What theoretical limits govern context inference once explicit task identifiers are removed?

The short answer is that MMALS should treat memory as a distributional and geometric object. A route is a point on a probability simplex. A route-function is a composition of route coefficients, host representations, fungal transformations, and downstream behavior. A memory trace is therefore not only a vector, but a law over route-function states induced by the data distribution and the learning context.

2 Classical MMALS notation

Let $x \in \mathcal{X}$ be an input, t be a learning context, and M the number of specialist hosts. Each host produces a latent representation

$$g_i(x; \theta_i) \in \mathbb{R}^d, \quad i = 1, \dots, M. \quad (1)$$

The fungal medium transforms host outputs,

$$\tilde{z}_i(x) = T_i(g_i(x; \theta_i); \psi_i), \quad (2)$$

and computes input/context-conditioned route coefficients

$$r_i(x, t) = \frac{\exp(a_i(x, t)/\tau_r)}{\sum_{j=1}^M \exp(a_j(x, t)/\tau_r)}, \quad r(x, t) \in \Delta^{M-1}. \quad (3)$$

The mediated latent exchange is

$$z_f(x, t) = \sum_{i=1}^M r_i(x, t) T_i(g_i(x; \theta_i); \psi_i). \quad (4)$$

A head or readout then produces a predictive distribution

$$p(y \mid x, t) = \text{softmax}(h_t(z_f(x, t))). \quad (5)$$

In early task-incremental prototypes, t contains an explicit task identifier. In the v1.0 direction, t must become a latent inferred context rather than a given label. This change is not cosmetic. It moves MMALS from supervised routing by task identity toward context inference under uncertainty.

2.1 The v0.9 non-implication

MMALS v0.9 introduced a simple but decisive non-implication. Suppose the route is unchanged:

$$r_{\text{current}}(x, t) \approx r_{\text{teacher}}(x, t). \quad (6)$$

This does not imply that

$$z_{f, \text{current}}(x, t) \approx z_{f, \text{teacher}}(x, t), \quad (7)$$

because the host representations g_i or transformations T_i may have changed. In the counterexample reported in the v0.9 notebook, $r_{\text{old}} = r_{\text{new}} = [1, 0, 0]$, yet the transformed first-host signal changes, giving $D_r = 0$ but $D_z > 0$ [6].

Proposition 1 (Route preservation is insufficient). *There exist MMALS states with identical route vectors and different mediated latent functions.*

Sketch. Let $r = [1, 0, 0]$ in both the teacher and current model. If $T_1^*(g_1^*(x)) \neq T_1(g_1(x))$, then Equation (4) gives $z_f^*(x, t) \neq z_f(x, t)$ despite identical routing. Therefore a metric on r alone cannot certify memory preservation. $\square$

This proposition explains why the mathematical backbone cannot stop at graph topology. The memory object must include the function computed through the route.

3 From point drift to law drift

Fréchet’s 1957 paper on the distance between two probability laws is relevant because MMALS memory is induced by data. For an old context τ , the teacher does not store one value of z_f ; it induces a distribution over $z_f(X, \tau)$ when X is sampled from the context data distribution. Fréchet explicitly distinguished a distance between random variables from a distance between their laws and emphasized that a distance between laws must be determined by the marginal distributions rather than by one arbitrary joint realization [3]. This is exactly the shift MMALS needs: compare memory laws, not only paired point values.

Definition 1 (Functional memory law). *For a context τ and a model state Θ , define the functional memory law*

$$\mu_\tau^{z, \Theta} = \mathcal{L}_{X \sim P_\tau}(z_f^\Theta(X, \tau)), \quad (8)$$

where $\mathcal{L}$ denotes the probability law of the mediated latent exchange. Similarly, define the route law

$$\mu_\tau^{r, \Theta} = \mathcal{L}_{X \sim P_\tau}(r^\Theta(X, \tau)), \quad (9)$$

and the behavioral law

$$\mu_\tau^{y, \Theta} = \mathcal{L}_{X \sim P_\tau}(p^\Theta(\cdot \mid X, \tau)). \quad (10)$$

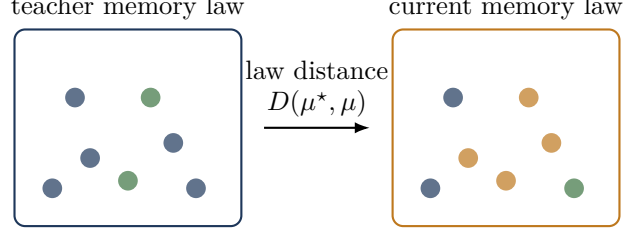


Figure 2: The law-level view: MMALS should compare distributions of functional memories, not only individual routes.

The v0.9 metrics are pointwise expectations:

$$D_r = \mathbb{E}_{x,t} \|r_{\text{current}}(x, t) - r_{\text{teacher}}(x, t)\|_2^2, \quad (11)$$

$$D_z = \mathbb{E}_{x,t} \|z_{f,\text{current}}(x, t) - z_{f,\text{teacher}}(x, t)\|_2^2, \quad (12)$$

$$D_y = \mathbb{E}_{x,t} D_{\text{KL}}(p_{\text{teacher}}(y | x, t) \| p_{\text{current}}(y | x, t)). \quad (13)$$

A law-level upgrade is

$$D_r^{\text{law}} = D(\mu_r^{r,\Theta^*}, \mu_r^{r,\Theta}), \quad (14)$$

$$D_z^{\text{law}} = D(\mu_r^{z,\Theta^*}, \mu_r^{z,\Theta}), \quad (15)$$

$$D_y^{\text{law}} = D(\mu_r^{y,\Theta^*}, \mu_r^{y,\Theta}), \quad (16)$$

where D is a distance or divergence between probability laws.

3.1 The jar-of-marbles analogy

For a 12-year-old, a law is like a jar of marbles. A single marble can look similar while the jar composition changes. A route vector is one marble; a route law is the jar. Fréchet’s insight allows MMALS to ask whether the whole jar of memory has changed.

4 Transport memory: how much work moves old memory into new memory?

Fréchet’s law-distance viewpoint naturally connects to optimal transport. If μ^* is the old functional memory law and μ the current law, then one may define a transport-style drift

$$\mathcal{T}_c(\mu^*, \mu) = \inf_{\gamma \in \Pi(\mu^*, \mu)} \int c(u, v) d\gamma(u, v), \quad (17)$$

where $\Pi(\mu^*, \mu)$ is the set of couplings with marginals μ^* and μ , and $c(u, v)$ is a cost for transforming one route-function state into another. Barbaresco’s Fréchet lecture slides explicitly place Fréchet–Lévy distance alongside optimal transport and compare it with Wasserstein-type distances [2]. Modern optimal transport provides the computational vocabulary for Equation (17) [10].

For MMALS, the cost can be designed at several levels:

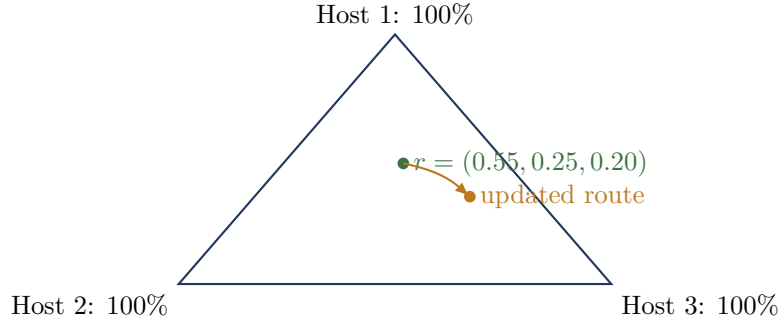
$$c_r(u, v) = \|u - v\|_2^2 \quad \text{route-only cost,} \quad (18)$$

$$c_z(u, v) = \|u - v\|_2^2 \quad \text{latent functional cost,} \quad (19)$$

$$c_y(u, v) = D_{\text{KL}}(u \| v) \quad \text{behavioral cost,} \quad (20)$$

$$c_{\text{hybrid}} = \alpha c_r + \beta c_z + \gamma c_y + \delta c_{\text{energy}}. \quad (21)$$

The sand-pile analogy is precise enough to be useful. If the old memory distribution is a sand pile and the new memory distribution is another sand pile, transport memory asks how much work is needed to reshape one into the other. Small work means stable memory. Large work means drift. Very large work may predict forgetting.



The probability playground: route updates must remain inside the simplex. Information geometry provides the correct local notion of distance and steepest descent.

Figure 3: Route coefficients live on a probability simplex. Geometry-MMALS should update them using manifold-aware methods rather than treating them as unconstrained Euclidean coordinates.

5 Information geometry: the route simplex is not flat Euclidean space

MMALS route coefficients satisfy

$$r(x, t) \in \Delta^{M-1} = \{r \in \mathbb{R}_{\geq 0}^M : \sum_i r_i = 1\}. \quad (22)$$

This constraint means routing does not live in unconstrained Euclidean space. Updating route logits with ordinary gradients may be computationally convenient, but it hides the geometry of probability distributions. Information geometry treats families of probability distributions as manifolds endowed with the Fisher metric [1]. Barbaresco’s document frames Fréchet’s work on distinguished densities, the Fréchet–Darmois bound, the Clairaut–Legendre transform, and the Hessian interpretation of the inverse Fisher matrix as a precursor to modern information geometry [2].

5.1 Natural-gradient route adaptation

Let θ denote route parameters, such as logits or natural parameters of an exponential family. A natural-gradient update takes the form

$$\theta_{k+1} = \theta_k - \eta I(\theta_k)^{-1} \nabla_{\theta} \mathcal{L}(\theta_k), \quad (23)$$

where $I(\theta)$ is the Fisher information matrix. The practical claim is not that all MMALS implementations must use full Fisher inversion. The claim is that geometry gives a principled direction for stabilizing route updates and measuring drift in a way compatible with probability constraints.

5.2 Exponential-family route memory

One can model routes as an exponential family:

$$p_{\theta}(r \mid x) = \exp(\langle \theta, T(r, x) \rangle - A(\theta)), \quad (24)$$

where $A(\theta)$ is a log-partition function. Then

$$\nabla^2 A(\theta) = I(\theta), \quad (25)$$

which makes the Hessian/Fisher geometry explicit. In this view, MMALS objects map cleanly to information-geometric objects:

Table 1: MMALS objects as information-geometric objects.

MMALS object	Information-geometric interpretation
Route logits	Natural parameters
Average route usage	Expectation parameters
Carbon cost	Entropy or convex potential regularization
Mutualistic gain	Utility constraint on used components
Functional memory drift	Geodesic, divergence, or transport drift between laws
Context uncertainty	Fisher information and estimator variance

6 Fréchet–Darmois bounds and the limit of context inference

MMALS v1.0 removes explicit task identity. The model must infer context from input, hidden state, route dynamics, and uncertainty signals. This suggests a statistical estimation problem. Let c denote a latent context and let $\hat{c}(X)$ be an estimator. Under standard regularity conditions, Cramér–Rao-type bounds state that the covariance of any unbiased estimator is lower-bounded by the inverse Fisher information:

$$\text{Cov}(\hat{c}) \succeq I(c)^{-1}. \quad (26)$$

Barbaresco’s source argues that the bound commonly called Cramér–Rao has a historical Fréchet–Darmois lineage, with Fréchet addressing the monivariate case and Darmois the multivariate case [2]. For MMALS, the historical naming is less important than the engineering implication:

If the signals available to the fungal medium contain little Fisher information about context, then context inference is intrinsically uncertain. The model should not behave as if it knows the context perfectly.

This motivates mixed, soft, or distributional memory states. It also provides an honest guardrail against overclaiming: bad context inference may not be an architecture bug; it may reflect insufficient statistical information.

7 Quantum-inspired functional memory as a later branch

The MMALS-QI note proposes a classical, quantum-inspired extension in which functional memory is represented by density operators over route-function states [8]. This is naturally downstream of the geometry program. A route vector lies in a simplex. A density operator lies in the positive semidefinite trace-one cone:

$$\rho \succeq 0, \quad \text{Tr}(\rho) = 1. \quad (27)$$

The simplex is recovered as the diagonal subset:

$$\rho_R(x, t) = \text{diag}(r_1(x, t), \dots, r_M(x, t)). \quad (28)$$

Thus MMALS-QI is not a rejection of classical MMALS. It is a generalization in which context uncertainty can be represented as a mixed state over possible route-function compositions.

A minimal state-memory loss is

$$\mathcal{L}_{\text{QI}} = \mathcal{L}_{\text{task}} + \lambda_y \mathcal{L}_{\text{distill}} + \lambda_z \|z_f^* - z_f\|_2^2 + \lambda_\rho D_\rho(\rho_\tau^*, \rho_\tau) + \lambda_S S(\rho_\tau) + \lambda_C C_{\text{metabolic}}. \quad (29)$$

Candidate metrics include fidelity, entropy, purity, trace distance, and Frobenius proxies [8]. The falsifiable hypothesis is that such state metrics should matter mainly in task-free, class-incremental, overlapping, or noisy-boundary regimes where a single route identity is insufficient.

Remark 1 (No quantum advantage claim). *All objects above can be implemented classically as matrices and differentiable losses. The value is representational and diagnostic, not a claim that MMALS is a quantum computer.*

8 Pedagogical compression: how to explain the backbone to a 12-year-old

A useful research idea should survive compression. The following comparisons preserve the essential mathematics without hiding the structure.

Table 2: Child-friendly analogies and their formal meaning.

Hard concept	12-year-old image	MMALS meaning
Route	Tunnel under a forest	Which host contributes to the answer
Functional memory	Meaning of the note in the tunnel	What the route actually computes
Fréchet law distance	Comparing two jars of marbles	Did the distribution of memories change?
Optimal transport	Moving one sand pile into another	How much work transforms old memory into new memory?
Fisher geometry	Probability playground	Route probabilities must stay valid and sum to one
Fréchet–Darmois bound	Detective with blurry clues	There is a limit to how well context can be inferred
Mixed-state memory	Several possible stories at once	Memory under ambiguous or overlapping contexts

The one-sentence child explanation is:

MMALS is like a forest of smart trees connected by underground fungus; the mathematical backbone is the set of maps and measuring tools that tells whether the fungus is preserving the useful meaning of messages, not only keeping the same tunnels.

9 Experimental roadmap

The mathematical backbone should be tested in stages, not asserted rhetorically.

9.1 Stage A: keep v0.11 as the comparative anchor

Prototype v0.11 is a comparative evidence harness rather than a new architecture. It asks whether MMALS provides a real continual-learning advantage compared with known baselines, including fine-tuning, joint upper bound, EWC, LwF, experience replay, PNN, sparse MoE, standard backbones, and MMALS variants [7]. This remains the anchor. Geometry should not be used to explain away weak empirical results.

9.2 Stage B: add law-level diagnostics without changing the model

Before modifying the architecture, compute:

$$D_r^{\text{law}}, \quad D_z^{\text{law}}, \quad D_y^{\text{law}}, \quad \mathcal{T}_c(\mu^*, \mu). \quad (30)$$

The question is diagnostic: do distributional drift metrics correlate with forgetting better than route-vector drift?

9.3 Stage C: Geometry-MMALS

Move route adaptation from unconstrained logit updates toward manifold-aware updates. Possible implementations include entropy-regularized natural gradients, mirror descent on the simplex, Riemannian trust-region updates, and Fisher-aware consolidation penalties.

9.4 Stage D: task-free and class-incremental MMALS

Remove explicit task identifiers. Context must be inferred. This is where Fréchet–Darmois/Fisher bounds become operational: the system should estimate not only a context, but also the confidence and identifiability of that context.

9.5 Stage E: MMALS-QI only after ambiguity is real

Density-state memory should not be expected to dominate in easy task-incremental settings. It becomes meaningful when contexts overlap, labels are ambiguous, or several functional routes remain plausible.

10 Falsification criteria

The mathematical backbone is useful only if it creates stricter tests. The following outcomes should weaken or delay the geometry branch:

1. Law-level drift metrics do not correlate with forgetting, output drift, or latent drift.
2. Transport memory merely duplicates MSE drift at higher computational cost.
3. Natural-gradient or mirror-descent routing improves neither stability nor interpretability.
4. Context Fisher-information estimates do not predict task-free routing uncertainty.
5. Density-state memory improves only by increasing parameter count or replay memory.
6. MMALS-QI collapses to diagonal routing in all meaningful ambiguous-context benchmarks.

A strong positive result would be:

In task-free or ambiguous-context continual learning, distributional and geometric memory metrics predict forgetting better than route-vector drift, and geometry-aware routing improves stability/plasticity at comparable parameter and memory cost.

A weaker but still valuable result would be:

The geometry branch does not improve final accuracy, but it explains when and why functional memory fails more clearly than existing diagnostics.

11 Discussion: what this contributes to MMALS

The mathematical backbone does not make MMALS true. It makes MMALS more falsifiable. It converts a biological analogy into a sequence of objects: route functions, memory laws, transport costs, Fisher metrics, inference bounds, and state representations. This shifts the research question from “Does a mycelium-like model sound plausible?” to four testable questions:

1. Does preserving functional memory laws reduce forgetting?
2. Do geometry-aware route updates stabilize the stability/plasticity trade-off?

3. Does Fisher-style context identifiability predict task-free routing failures?
4. Does mixed-state route memory help only when context is genuinely ambiguous?

This is also why the 12-year-old explanation is not a toy add-on. The forest–tunnel–jar–sand–playground analogies expose the exact conceptual layers that the formal model must preserve. The tunnel is the route. The note is the function. The jar is the law. The sand movement is transport. The playground is the simplex. The detective limit is Fisher information. The mathematical backbone is the system of rules that prevents the story from becoming vague.

12 Conclusion

MMALS began as a mycelium-inspired continual-learning architecture: hosts exchange transformed latent signals through an adaptive fungal medium. The v0.9 lesson forced a conceptual upgrade: memory is not the route; memory is the function computed through the route. Fréchet’s distance between probability laws suggests the next upgrade: memory is not only a pointwise function, but a distribution of functional route states. Optimal transport turns that distributional comparison into a cost of memory deformation. Information geometry gives a principled metric for route movement on probability simplexes. Fréchet–Darmois/Fisher bounds explain why context inference has unavoidable limits. MMALS-QI, if pursued later, generalizes probabilistic routing into mixed-state functional memory under ambiguity.

The conservative research claim is therefore:

MMALS becomes scientifically stronger when its mycelial analogy is grounded in distributional memory, information geometry, and uncertainty-aware context inference; these tools do not prove MMALS, but they define sharper diagnostics, cleaner falsification criteria, and a credible path from v0.x prototypes to task-free Geometry-MMALS.

A Minimal implementation checklist

1. Store teacher samples of $r^*(x, \tau)$, $z_f^*(x, \tau)$, and $p^*(y \mid x, \tau)$ for old contexts.
2. Estimate empirical laws $\hat{\mu}_\tau^r$, $\hat{\mu}_\tau^z$, and $\hat{\mu}_\tau^y$ using held-out probes.
3. Compute pointwise metrics D_r, D_z, D_y and law-level metrics $D_r^{\text{law}}, D_z^{\text{law}}, D_y^{\text{law}}$.
4. Add a transport-memory diagnostic using Sinkhorn or sliced Wasserstein approximations.
5. Correlate each diagnostic with final forgetting and per-task degradation.
6. Only after diagnostic value is established, test route optimization variants: mirror descent, natural gradient, or Fisher-aware consolidation.
7. For task-free learning, estimate context uncertainty and compare it with route entropy and error.
8. For MMALS-QI, begin with real-valued PSD matrices $\rho = UU^\top / \text{Tr}(UU^\top)$ and use Frobenius proxies before implementing expensive fidelity metrics.

B Suggested figures for a future carousel

1. Forest system: hosts as trees, fungal medium as underground exchange layer.
2. Same tunnel, different note: route preservation does not imply functional preservation.

3. Two jars of marbles: probability law drift.
4. Two sand piles: transport cost between memory distributions.
5. Probability triangle: route simplex and Fisher geometry.
6. Detective with blurry clues: Fréchet–Darmois/Fisher limit on context inference.
7. Mixed memory state: several possible route-functions under uncertain context.

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